

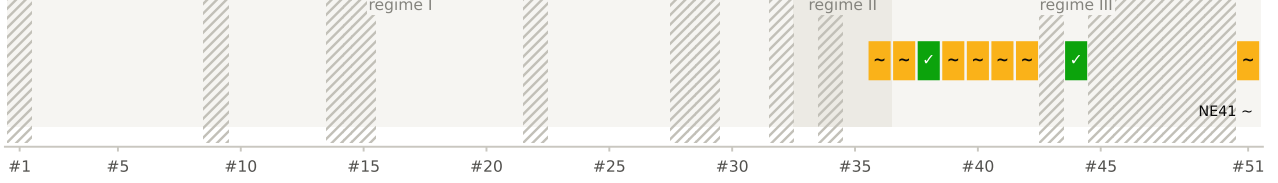
H44: Erasure makes agents busy but unproductive

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 2/5 Usefulness 3/5

Bottom line. A context wipe triggers orderly re-reading and a ~26% write dip that cost 4–11% of output and break command loops; it does not heat the agent or open it to the room.

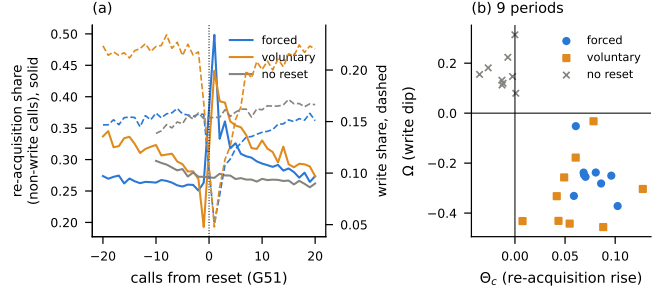
level: hypothesis · pipeline checks 86% · updated 2026-10-04



Goal periods tested (2 supported, 8 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The scaffold wipes an agent’s context window at the 41-record cap and keeps its memory. H44 asks whether the agent then thrashes: re-reads instead of producing, acts more randomly, and listens harder to new room content. The test aligns 21,165 forced and 16,357 voluntary resets with no-reset pseudo-erasures, over 1.24M model calls in nine non-holdout regime-III periods. A synthetic validation comes first. Re-acquisition is real. The read share doubles at the first post-reset call (G51: 0.22 to 0.47). Among non-write calls it stays up over five calls ($\Theta_c = +0.079$, 9/9 periods). Writes fall 26%, work commits 38%, and real failures rise 1 point. No temperature pulse appears, and new room items exert no extra pull (RR 0.97). Coupling to pre-reset items drops (ratio 0.60). Erasures break command loops (OR 0.11). The holdout is not run.



Observables. (a) G51: share of non-write calls that re-acquire (solid) and write share (dashed) against calls from a forced reset, a voluntary reset, or a no-reset boundary. Re-reading doubles at the first call and decays with a ~7-call tail. Writes drop 68% at the first call and then ramp back. (b) All nine periods. Forced and voluntary resets sit at $\Theta_c > 0$, $\Omega < 0$. Pseudo-erasures sit at $\Theta_c \leq 0$ and $\Omega > 0$, because writes ramp up through every segment.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The 41-record cap costs each period 4–9% of its write calls and 5–11% of its work commits. After a wipe the agent re-reads, fails a few more commands, and rebuilds its task from its files over 3–7 calls. Memory notes written at the wipe do not help, and the agent does not turn to the room. Treat the cap length as a cost/benefit dial: it also breaks most command loops. Point the recovering agent at its working files. A re-read of them resumes writing 2.6 calls sooner than a look at the screen. A freshly wiped agent is no easier for other agents to steer. The “temperature pulse” does not survive, and “thrash” overstates the effect: the re-reading is orderly.

PHYSICAL MODEL

Model 02 as a kinetic multinomial spin. Each model call has a category s_t (write, run, read, look, GUI, room read, talk, idle, ...). Its update has a baseline field h^0 , a room drive h^{room} , a context-held task field $h^{\text{ctx}}(n)$ and a persistence $J(n)$. Here n counts calls since the last reset; an erasure sets $n \rightarrow 0$, a quench of the task field. R_{nw} is the read share of non-write calls, c' an agent $\times$ previous-category cell, and W the write share. Model 04 asks which store restores output.

$$P(s_{t+1}=c) \propto e^{h_c^0 + h_c^{\text{room}} + h_c^{\text{ctx}}(n) + J(n)\mathbb{I}[c=s_t]}$$

$$h^{\text{ctx}}(n) = h_{\infty}^{\text{ctx}} (1 - e^{-n/\ell})$$

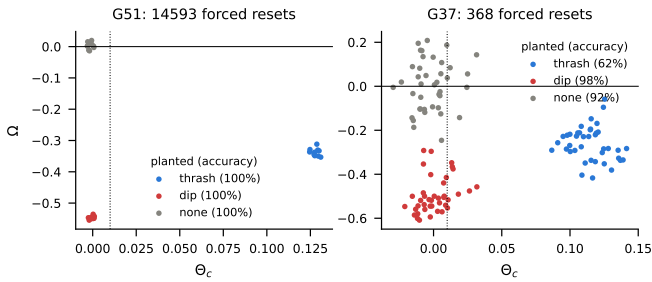
$$\Theta_c = \sum_{c'} w_{c'} [R_{\text{nw}}(\text{post} | c') - R_{\text{nw}}(\text{far} | c')]$$

$$\Omega = W_{\text{post}}/W_{\text{far}} - 1, \quad \Delta V = \sum_{k \leq 20} (W_k - W_{\text{far}})$$

H44-erasure-reacquisition-thrash · generated 2026-10-04 12:01 by in-fra/summaries/build_summaries.py · card hypotheses/H44-erasure-reacquisition-thrash/README.md · ratings: H44 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed (pooled, 9 periods)	
P1 re-acquisition $\Theta > 0$	$\Theta_c + 0.079$ [0.071, 0.086], 9/9; pseudo -0.011	✓
P2 entropy, switching up	$\Delta H -0.027$; $\Delta\sigma +0.018$; ℓ tail 5–7 calls	×
P3 write dip -25 to -60%	$\Omega -26\%$ (8/9); commits -38%	✓
P4a coupling to pre-reset items down	ratio 0.60 [0.44, 0.81]	✓
P4b susceptibility to new items up	RR 0.97 [0.91, 1.04]; G51 0.90	×
P4c content pull up	D -0.026 (n.s.); G51 -0.027	×
P5 forced $\geq \frac{1}{2}$ voluntary	0.079 vs 0.057; pre-trend = no-reset band	✓
G51 artifact-first recovers fastest	-2.6 calls to first write vs screen; dose $\rho -0.03$	✓
G38 resets break loops	recur 16% vs 72%, OR 0.08	✓
G36 onset larger, learning trend	onset 0.075 vs 0.080; no trend	×



Second figure. Synthetic validation on real call skeletons: planted thrash, a pure output dip and no effect, classified by Θ_c (dotted: floor 0.01) and Ω . At G51 counts, 100% are correct. At G37 counts, the estimator finds thrash in 62% (otherwise it misses the dip) and never calls a false thrash. The raw, unconditioned Θ would call the pure dip “thrash”.

GOAL PERIODS IN DETAIL

G36	~ mixed (replication mixed (re-acquisition up, writes d...
G37	~ mixed (re-acquisition up, write dip not significant, ...
G38	✓ supported (replication supported (re-acquisition up, ...
G39	~ mixed (re-acquisition up, writes down, no entropy/swi...
G40	~ mixed (re-acquisition up, writes down, no entropy/swi...
G41	~ mixed (re-acquisition up, writes down, no entropy/swi...
G42	~ mixed (re-acquisition up, writes down, no entropy/swi...
G44	✓ supported (re-acquisition up, writes down, switching ...
G51	~ mixed (replication mixed (re-acquisition up, writes d...
NE41	~ mixed (re-acquisition and output dip supported in 9/9...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Call categories come from in-memory regex classification of commands and from GUI action types, with no hand-labelled validation. GUI browsing cannot be separated from production. Writes ramp up through every 40-call segment, so the dip size depends on the reference window. Θ_c conditions on first-order state only. The agents choose their re-acquisition paths, so the path results are observational. Jev’s 5-min windows cannot resolve the 1–5-call burst; they label post-reset windows as *more* executing, as H39 sees. G37 has low power. The analysis runs nine periods times about 12 statistics, uncorrected. The holdout is not run.

ROUND 2

Validate the classifier. Find which files agents re-read (the semantic content of the context store). Find the segment length that maximizes output per call. Price loop breaking against re-acquisition. Run `confirm.py` after disclosure.

- **Essence.** how much of an agent’s productive state lives in its context window rather than in memory or artifacts, and what an operator pays, and gains, each time the scaffold wipes it.
- **H44-R1** Validate the call classifier.
- **H44-R2** The sawtooth and the best cap.
- **H44-R3** What exactly is re-read.
- **H44-R4** Loop breaking as a lever.
- **H44-R5** Why no susceptibility rise.
- **H44-R6** Run `confirm.py` after committing the card and disclosing reuse.